Economics for Engineers (Humanities-II) (HSMC-301)

Course Details

Credit Points: 3

Syllabus Outline

Unit 1

- **Economic Decisions Making** – Overview, Problems, Role, Decision making process.
- **Engineering Costs & Estimation** – Fixed, Variable, Marginal & Average Costs, Sunk Costs, Opportunity Costs, Recurring and Nonrecurring Costs, Incremental Costs, Cash Costs vs Book Costs, Life-Cycle Costs
- **Types Of Estimates, Estimating Models** - Per-Unit Model, Segmenting Model, Cost Indexes, Power-Sizing Model, Improvement & Learning Curve, Benefits.

Unit 2

Cash Flow, Interest and Equivalence

Versity.onrender.com

- **Cash Flow** – Diagrams, Categories & Computation, Time Value of Money, Debt repayment, Nominal & Effective Interest
- **Cash Flow & Rate of Return Analysis** – Calculations, Treatment of Salvage Value, Annual Cash Flow Analysis, Analysis Periods; Internal Rate of Return, Calculating Rate of Return, Incremental Analysis; Best Alternative Choosing an Analysis Method, Future Worth Analysis, Benefit-Cost Ratio Analysis, Sensitivity and Breakeven Analysis
- **Economic Analysis in The Public Sector** -Quantifying and Valuing Benefits & drawbacks.

Unit 3

- **Inflation and Price Change** – Definition, Effects, Causes, Price Change with Indexes, Types of Indexes, Composite vs Commodity Indexes, Use of Price Indexes in Engineering Economic Analysis, Cash Flows that inflate at different Rates
- **Present Worth Analysis** - End-Of-Year Convention, Viewpoint of Economic Analysis Studies, Borrowed Money Viewpoint, Effect of Inflation & Deflation, Taxes, Economic Criteria, Applying Present Worth Techniques, Multiple Alternatives
- **Uncertainty In Future Events** - Estimates and Their Use in Economic Analysis, Range of Estimates, Probability, Joint Probability Distributions, Expected Value, Economic Decision Trees, Risk, Risk vs Return, Simulation, Real Options.

Unit 4

- **Depreciation** - Basic Aspects, Deterioration & Obsolescence, Depreciation and Expenses, Types of Property, Depreciation Calculation Fundamentals, Depreciation and Capital Allowance Methods, Straight-Line Depreciation, Declining Balance Depreciation, Common Elements of Tax Regulations for Depreciation and Capital Allowances
- **Replacement Analysis** - Replacement Analysis Decision Map, Minimum Cost Life of a New Asset, Marginal Cost, Minimum Cost Life Problems
- **Accounting** – Function, Balance Sheet, Income Statement, Financial Ratios Capital Transactions, Cost Accounting, Direct and Indirect Costs, Indirect Cost Allocation.

Versity.onrender.com